

Avi Gobrin

✉ avichaigobrin@gmail.com | ☎ (416) 670-2948 | [in linkedin.com/in/avigobrin](https://www.linkedin.com/in/avigobrin) | github.com/Avi-Gobrin | avi-gobrin.github.io

Education

University of Toronto

2023 – Apr 2027

Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor

Toronto, ON

Relevant Coursework: Algorithm Design & Analysis, Data Structures, Software Design, Theory of Computation

Experience

Nextdoor

May 2026 – Aug 2026

Machine Learning Intern

San Francisco, CA

- Increased growth and weekly active users by launching Best Places for Families, an SEO ranking product covering ~300,000 US neighbourhoods, building every layer end to end: Airflow pipelines ingesting internal and Census/ACS data, a backend scoring service, and a Postgres/GraphQL serving path with a React frontend.
- Designed the scoring methodology by comparing 4 models predicting family-friendliness scores for the 197,556 of ~300,000 neighbourhoods with sufficient data; screened 33 metrics down to 8 features.
- Validated ranking quality before launch through an evaluation harness: a preregistered neighbourhood adversarial test set with pairwise Bradley-Terry judgments, cross-checked by both LLM and manual review.
- Made the system self-sustaining with scheduled jobs, tests, and monitoring, and built Databricks tooling that lets non-technical stakeholders retune feature weights and see how any city ranks.

Shopify

May – Aug 2025

Software Engineering Intern

Toronto, ON

- Increased merchant activation and retention by building a full-stack partner-attach system (Ruby on Rails, Calendly API) that screens each new merchant and books qualifying ones into guided onboarding with a vetted advisor. Routing over 100 merchants per day into the partner attach flow.
- Brought Shopify's entire German merchant base into VAT compliance in the 1.5 months from regulatory notice to launch. Reconciled what data Shopify already held against what merchants had to supply, integrated a difficult government tax API, and built a Rails intake form with an email flow that returns each merchant's compliance PDF.
- Cleared reliability-impacting technical debt across adjacent teams and codebases, increasing engineering output and velocity.

Research

University of Toronto, Department of Statistical Sciences

Expected: Fall 2026

Incoming Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute)

Toronto, ON

- Investigating a unifying rank-reduction framework linking in-context learning and fine-tuning as low-rank weight updates, connecting recent transformer interpretability work to classical multivariate statistics.

Projects

GPT from Scratch *Personal Project (in progress)* [\[Repo\]](#)

2026

- Building a character-level GPT in pure NumPy, growing one codebase from a bigram baseline into a full Transformer: tokenization, stable softmax, multi-head causal self-attention, LayerNorm, residual connections, Adam, and temperature/top-k sampling; every forward and backward pass derived on paper before being coded, no autograd.

Autonomous Planning & School Knowledge Agents *Personal Project*

2026

- Cut daily planning from about 30 minutes to under 2 with an autonomous planning agent (Claude Code, Obsidian) that runs at the start and end of each day, reviewing what was completed, writing structured notes on the day, and regenerating a time-blocked calendar for the next day from a parsed task file and a traffic-light priority scheme.
- Cut note-cleanup and lookup time an estimated 40% with a companion school-workflow agent that auto-formats lecture and problem-set notes into a cross-linked knowledge base of theorem references, surfacing relevant prior material to accelerate homework and exam preparation.

Plantarium *6-person team, Java* [\[Repo\]](#)

2024

- Collaborated on an AI plant-identification desktop app (Java, Maven) covering photo upload to species identification, public/private galleries, likes, and notes.
- Owned the account-management flow, including secure account deletion with full data wipe.

Technical Skills

Languages: Python, Java, JavaScript, SQL, Bash, HTML, CSS, R

Systems & Web: Ruby on Rails, Postgres, GraphQL, React, Node.js, Flask, REST APIs, Airflow, Databricks, Linux/Unix

ML & Data: PyTorch, scikit-learn, NumPy, Pandas, SciPy, statsmodels; neural nets and backprop implemented from scratch

Tools: Git, GitHub, Claude Code, Cursor, Claude, LLM agents, Jupyter, Matplotlib, Seaborn, Maven, Excel, L^AT_EX